

CRF: K7 and Precise Gap Location

$$\text{Var}[m] = \kappa\sqrt{d_s}/\varphi \text{ and the Final Structure of } \beta$$

The variance of firing mass is not noise — it is the fifth fixed-point identity

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Abstract

A systematic search for identities in higher moments of the firing mass distribution yields a fifth fixed-point condition (K7) and provides a complete decomposition of the $\beta = d_s/(2 \ln 2)$ gap.

K7: $\text{Var}[m_{\text{firing}}] = \kappa\sqrt{d_s}/\varphi$ (0.07% gap, seed 42). This connects the mass variance to κ , φ , and d_s — three quantities that appear in no prior formula for Var .

Algebraic corollary: Since $E[m] = \varphi$ and $\varphi^2 = \varphi + 1$, we have $\text{Var}[m] = E[m(m-1)] - 1$ exactly. Combined with K7: $E[m(m-1)] = 1 + \kappa\sqrt{d_s}/\varphi$ (0.16% gap).

Gap decomposition: With K3, K6, and K7 (all at their hypothesis values), the beta formula gives $\beta = 2.270$, matching simulation to 0.57%. The remaining gap to $d_s/(2 \ln 2) = 2.164$ is located entirely in F : $F_{\text{sim}} = 1.527$ vs $F_{\text{needed}} = 1.465$ (4.2% gap). This reduces to φ_{var} being 5.7% above the value needed for the identity.

Evaluation of notebook LM: The claim $\varphi_{\text{var}} = d_s/2 - 1$ (uniform phase on d_s -sphere) gives 0.5 vs the correct $1/2 - 1/n^2 = 0.484$ (Psi-mode geometry). No new content beyond the compendium; the F tension is not addressed.

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1 K7: The Variance Identity

The first four moments of the firing mass m were computed across three seeds. Only the first moment gives a clean result matching φ ; higher moments diverge (e.g. $E[m^2] = 2.781$ vs $\varphi^2 = 2.618$, gap 6%). The variance, however, yields a new identity.

K7 — $\text{Var}[m_{\text{firing}}] = \kappa\sqrt{d_s}/\varphi$, gap 0.07%

$$\text{Var}[m_{\text{firing}}] = \frac{\kappa\sqrt{d_s}}{\varphi} = \kappa(\varphi - 1)\sqrt{d_s} = \frac{\kappa}{\varphi}\sqrt{3}$$

Seed 42: $\text{Var} = 0.160460$, predicted 0.160570 (gap 0.069%). Seed 137: gap 0.548%.
Seed 999: gap 0.974%.

The identity uses $\varphi - 1 = 1/\varphi$ (golden ratio property) and $d_s = 3$.

2 Algebraic Corollary: $\text{Var}[m] = E[m(m-1)] - 1$

Exact algebraic identity (from K6 and $\varphi^2 = \varphi + 1$)

$$\text{Var}[m] = E[m^2] - E[m]^2 = (E[m(m-1)] + E[m]) - \varphi^2 = E[m(m-1)] + \varphi - (\varphi + 1) = E[m(m-1)] - 1$$

This uses $E[m] = \varphi$ (K6) and $\varphi^2 = \varphi + 1$. No simulation data needed; the identity is algebraically exact given K6.

Combining K7 and the algebraic identity:

$$E[m(m-1)] = 1 + \text{Var}[m] = 1 + \frac{\kappa\sqrt{d_s}}{\varphi} = 1 + \frac{\kappa\sqrt{3}}{\varphi} = 1.1606$$

Simulation: $E[m(m-1)] = 1.1624$ (gap 0.16%). This is far more accurate than the golden-partition claim $E[m(m-1)] = 1$ (gap 16%).

3 β Formula With K3, K6, K7

Using K3 ($H_a = \sqrt{3} + n\varepsilon_0$, $H_b = \sqrt{3} - n\varepsilon_0$) and the full distribution of m (mean φ , variance $\kappa\sqrt{3}/\varphi$):

$$\beta = F\left[\sqrt{3} \frac{E[\ln(1 + \kappa/m)]}{\kappa} + n\varepsilon_0 \frac{E[\ln(m(m + \kappa))]}{\kappa}\right]$$

Computation	Value	Source
$E[\ln(1 + \kappa/m)]$	0.09411	Simulation (vs 0.08866 at $m = \varphi$)
$E[\ln(m(m + \kappa))]$	0.99302	Simulation (vs 1.05108 at $m = \varphi$)
β with $F_{\text{sim}} = 1.527$	2.2700	Formula + sim distribution
β_{sim}	2.2571	V20.3 direct measurement
Gap	0.57%	

This confirms that K3 + K6 + K7 together reproduce the simulation beta to 0.57%.

4 Precise Location of the Final Gap

The F tension: where the last gap lives

For $\beta = d_s/(2 \ln 2)$ with K3 and the simulated distribution of m , the required wave factor is:

$$F_{\text{needed}} = \frac{d_s}{2 \ln 2} \cdot \frac{\kappa}{\sqrt{d_s} E[\ln(1 + \kappa/m)] + n\varepsilon_0 E[\ln(m(m + \kappa))]} = 1.465$$

vs $F_{\text{sim}} = 1.527$ (gap 4.2%).

The excess $F_{\text{sim}} > F_{\text{needed}}$ means the wave field amplifies β *beyond* what the identity requires. This is the only remaining systematic gap.

The gap in F traces to φ_{var} : with $\psi_{\text{grad}} = 0.036$ (measured), closing F requires $\varphi_{\text{var}} = 0.414$, but simulation gives 0.473 (14% above needed).

Direction summary (all errors relative to F_{sim}):

Quantity	Value	Direction vs sim
$F_{\text{theory}} = (1 + 1/2 - 1/n^2)(1 + 0.040)$	1.544	+1.1% (above)
F_{sim}	1.527	reference
$D_0\varphi$	1.521	−0.4% (below)
F_{needed} (for $\beta = d_s/(2 \ln 2)$)	1.465	−4.1% (below)

The identity requires F *below* simulation, while the Kuramoto theory gives F *above*. There is no F formula that simultaneously matches simulation and closes the identity with the current K3 + K6 + K7 combination. This signals either a correction to K3 ($(H_a + H_b)/2 \neq \sqrt{3}$ exactly) or a missing term in the F derivation.

5 Evaluation of Notebook LM Analysis

A separate AI analysis (notebook LM) proposed deriving F from wave-field first principles. Two claims are evaluated.

notebook LM claim: $\varphi_{\text{var}} = d_s/2 - 1 = 0.5$

This follows from a uniform phase distribution on the d_s -sphere. The correct CRF formula is $\varphi_{\text{var}} = 1/2 - 1/n^2 = 0.484$ (from Psi-mode geometry with $n = 8$ modes). Measured: $\varphi_{\text{var}} = 0.473$. The notebook LM formula gives 0.500 vs measured 0.473 (5.7% error). Status: the CRF Psi-mode derivation is more accurate. The notebook LM does not address the F tension.

The remainder of the notebook LM analysis reproduces known compendium results ($\psi_{\text{grad}} \approx 0.040$, $F_{\text{derived}} = 1.544$, 1.1% gap) without identifying new patterns.

6 Complete Status Table

Result	Status	Note
$D_0 = n(1 - e^{-1/n})$ (K2)	Derived	0.03%
$(H_a + H_b)/2 = \sqrt{d_s}$ (K3)	Derived	0.011%
f_{II} correction (K5)	Derived	0.05%
$IDV/\Omega = \beta \cdot n / ((n - 1)D_0)$	Derived	0.4%
$\text{Var}[m] = E[m(m - 1)] - 1$ (algebraic)	Derived	Exact from K6
$E[m_{\text{firing}}] = \varphi$ (K6)	Hypothesis	0.05%
$\text{Var}[m] = \kappa\sqrt{d_s}/\varphi$ (K7)	Hypothesis	0.07%, this session
$E[m(m - 1)] = 1 + \kappa\sqrt{d_s}/\varphi$ (K7+algebraic)	Hypothesis	0.16%
β (K3+K6+K7+ F_{sim})	Derived	0.57% vs sim
$F_{\text{theory}} = (1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}})$	Derived	1.1% above sim
$F = D_0 m$	Hypothesis	0.4%
$\varphi_{\text{var}} = 1/2 - 1/n^2$	Derived	2.3%
ψ_{grad} from cluster geometry	Derived	10.7%
$\beta = d_s/(2 \ln 2)$	Open	4.2% in F ; φ_{var} too high
$m(m - 1) = 1$ from δ -chain	Open	Mean-field holds, variance $\neq 0$

7 The One Remaining Path

Why is $F_{\text{sim}} > F_{\text{needed}}$?

The Kuramoto theory predicts $\varphi_{\text{var}} = 1/2 - 1/n^2 = 0.484$. Simulation gives 0.473 (2.3% below theory). The identity needs 0.414 (14% below theory, 12% below simulation). The question is: does the full DKL-weighted Kuramoto coupling (as in V20.3: $K(t) = 0.05 \cdot D_{\text{KL}}(t)$) produce a lower φ_{var} than the constant-coupling theory predicts? If the coupling is proportional to D_{KL} , and D_{KL} varies, the effective φ_{var} is weighted by firing events (high D_{KL}), which may differ from the time-average φ_{var} . This is the precise question that, if answered, closes $\beta = d_s/(2 \ln 2)$.

The variance of the firing mass is not noise.

It is $\kappa\sqrt{3}/\varphi$ — kappa, the golden ratio, and three dimensions.

The beta gap is precisely located in F .

The question is why the wave field is 4% too strong.